

Qn	Suggested Answer
1	
(a)(i)	$\Delta(m_1 + m_2) = \Delta m_1 + \Delta m_2$ $= (\pm 1) + (\pm 1)$ $= \pm 2 \text{ g}$
(a)(ii)	$\frac{\Delta T}{T} \times 100\% = \frac{1}{2} \left(\frac{\Delta m_1}{m_1} + \frac{\Delta m_2}{m_2} + \frac{\Delta k}{k} + \frac{\Delta(m_1 + m_2)}{m_1 + m_2} \right) \times 100\%$
	$\frac{\Delta T}{T} \times 100\% = \pm \frac{1}{2} \left(\frac{1}{500} + \frac{1}{1167} + \frac{0.2}{8.0} + \frac{2}{500 + 1167} \right) \times 100\%$
	$\frac{\Delta T}{T} \times 100\% = \pm 1.45$ $= \pm 1.5\% \text{ (2sf)}$
	Alternative solution using max/min method
	$T_{\max} = 2\pi \sqrt{\frac{(501 \times 10^{-3})(1168 \times 10^{-3})}{(7.8)[(501 \times 10^{-3}) + (1168 \times 10^{-3})]}}$ $= 1.332 \text{ s}$ $T_{\min} = 2\pi \sqrt{\frac{(499 \times 10^{-3})(1166 \times 10^{-3})}{(8.2)[(499 \times 10^{-3}) + (1166 \times 10^{-3})]}}$ $= 1.297 \text{ s}$
	$\frac{\Delta T}{T} \times 100\% = \frac{0.5(1.332 - 1.297)}{0.5(1.332 + 1.297)} \times 100\%$
	$\frac{\Delta T}{T} \times 100\% = \pm 1.33$ $= \pm 1.3\% \text{ (2sf)}$
(b)(i)	Uncertainty of T averaged from 20 oscillations $= \frac{\pm 0.4}{20}$ $= \pm 0.02 \text{ s}$
	Uncertainty of T if the time for one oscillation is recorded is $\pm 0.4 \text{ s}$. Hence, finding the average time from 20 oscillations will <u>reduce</u> the <u>uncertainty</u> of T .
(b)(ii)	The time recorded as the period would have been for half an oscillation. percentage error = $\frac{\text{wrong value} - \text{correct value}}{\text{correct value}} \times 100\%$ $= \frac{0.5T - T}{T} \times 100\%$
	percentage error = -50%
(b)(iii)	The human error in (ii) is a <u>systematic error</u> . Systematic errors cannot be reduced by finding the average time from multiple oscillations.

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2	